

DATA SCIENCE TOOLBOX PYTHON

PROJECT REPORT

(Project Semester January-April 2025)

ELECTRIC VEHICLE DASHBOARD ANALYSIS

Submitted by

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Programme and Section:- Btech Cse, K23GD

Course Code INT375

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Samit, bearing Registration no. 12320019 has completed INT 375 project titled, “**Analysis and Visualization of Electric Vehicle (EV) Adoption Trends in the U.S.**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Baljinder Kaur

School of Computer Science

Lovely Professional University

Phagwara, Punjab.

Date: 12-04-2025

DECLARATION

I, Samit, student of Computer Science under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-04-2025

Signature

Registration No. 12320019

Samit

Acknowledgement

I would like to express my sincere gratitude to my project guide, [Guide Name], for their invaluable support, insightful feedback, and constant encouragement throughout the project. I would also like to thank the faculty and staff of the Department of Computer Science and Engineering for their assistance. Finally, I extend my heartfelt thanks to my family and friends who supported me during this project.

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1. Introduction

The objective of this project is to analyze Electric Vehicle (EV) data through a visually interactive dashboard using Python libraries such as Matplotlib, Seaborn, and Pandas. It aims to derive meaningful insights from the dataset, such as EV distribution by model year, city, range, price, and type.

2. Source of Dataset

The dataset was collected from publicly available EV registration records, which include attributes such as Model Year, City, Make, Electric Range, Base MSRP, and EV Type. It was cleaned using Pandas for missing or inconsistent data.

3. EDA Process

Exploratory Data Analysis (EDA) was conducted to:

- Understand data distribution and outliers
- Handle missing values
- Generate frequency distributions
- Plot visualizations for each significant variable

The following libraries were used:

- pandas for data cleaning and manipulation
 - seaborn and matplotlib for visualization
-

4. Analysis on Dataset

i. Introduction

The analysis focuses on extracting key metrics and trends from the EV dataset, divided into two main dashboards: General Insights and Advanced Trends.

ii. General Description

- **Model Year Distribution:** Shows the count of EVs by manufacturing year.
- **Top 10 Cities:** Displays cities with the highest EV registrations.
- **Electric Range:** Distribution of how far EVs can travel on a full charge.
- **Base MSRP:** Histogram showing vehicle cost distribution.
- **EV Type Distribution:** Pie chart of Battery Electric vs Plug-in Hybrid.

iii. Specific Requirements, Functions, and Formulas

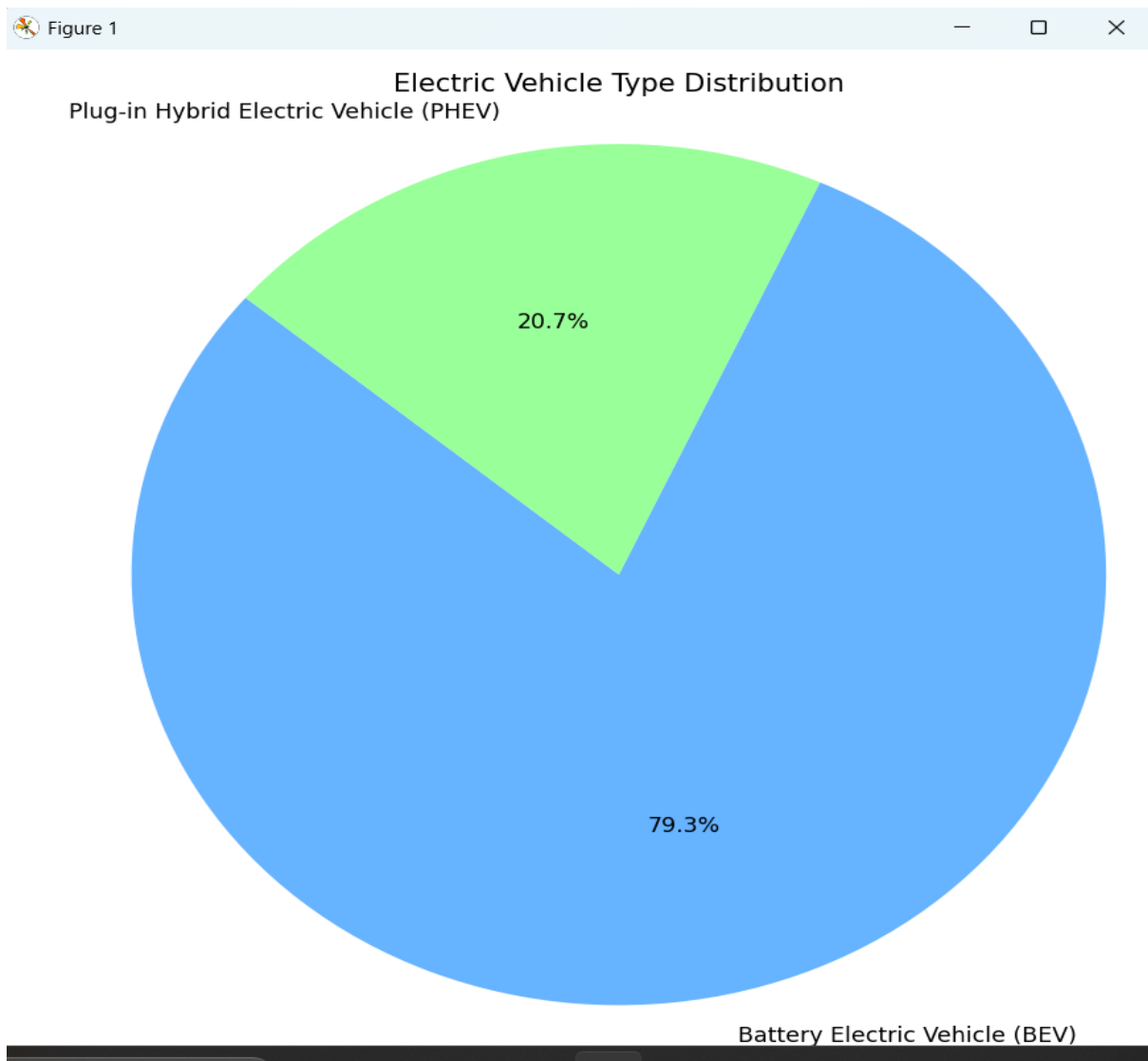
- `sns.countplot()` for categorical data frequency
- `df['Column'].dropna().plot.hist()` for numerical distributions
- `top_cities.plot(kind='barh')` for horizontal bar charts
- `type_counts.plot(kind='pie')` for pie chart
- `plt.subplots()` to combine plots into dashboards
- `plt.tight_layout()` and `fig.suptitle()` for spacing and titling

iv. Analysis Results

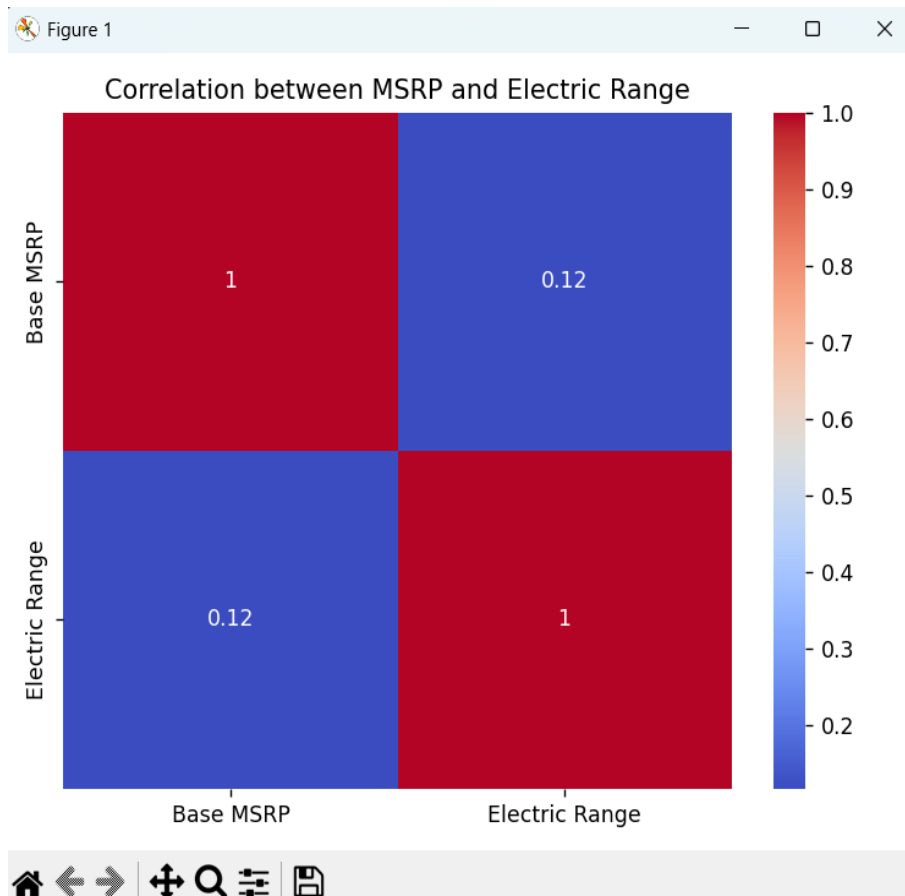
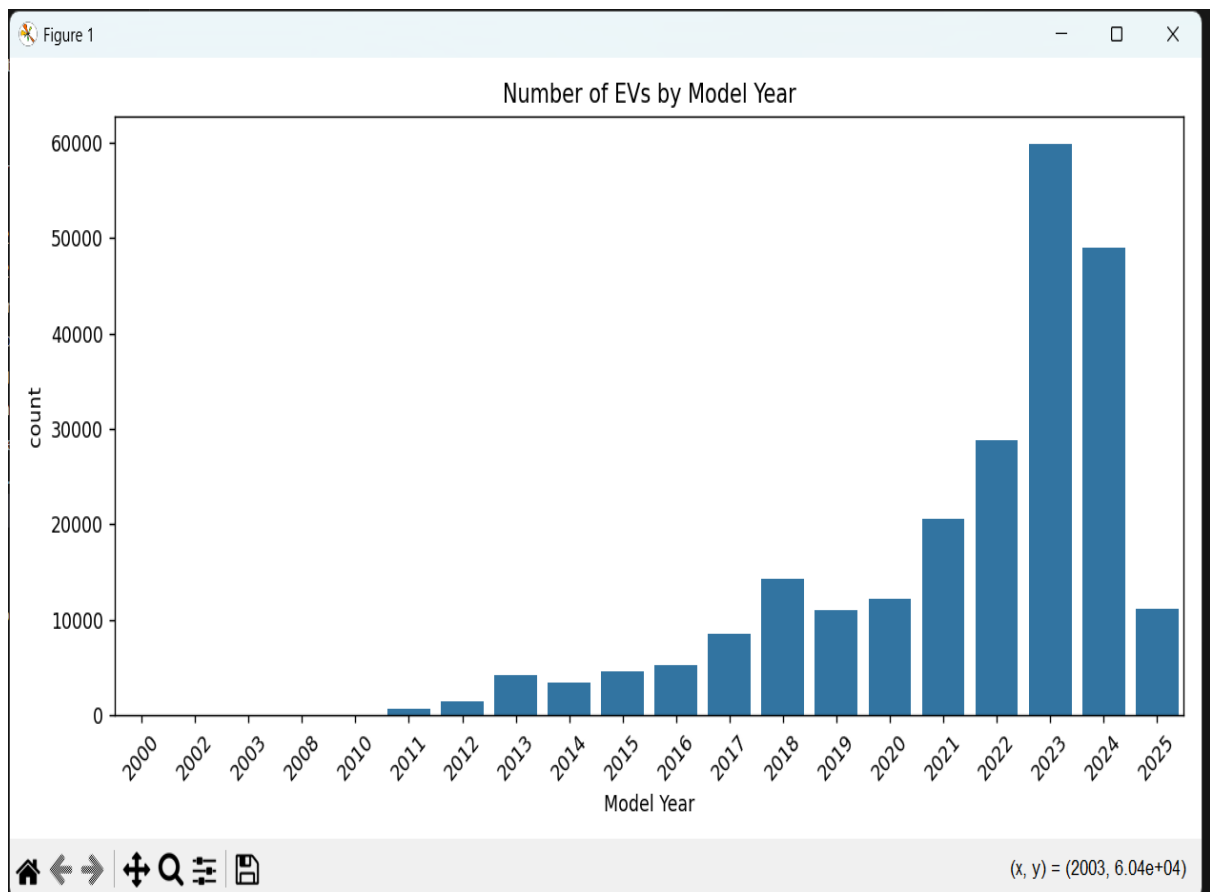
- Majority of EVs are recent (2020-2023)
- Urban areas like Seattle, Bellevue, and Redmond dominate EV adoption
- Most EVs have electric ranges between 75-300 miles
- Base MSRP is concentrated between \$30,000 - \$50,000
- 75% are Battery Electric Vehicles, 25% Plug-in Hybrid

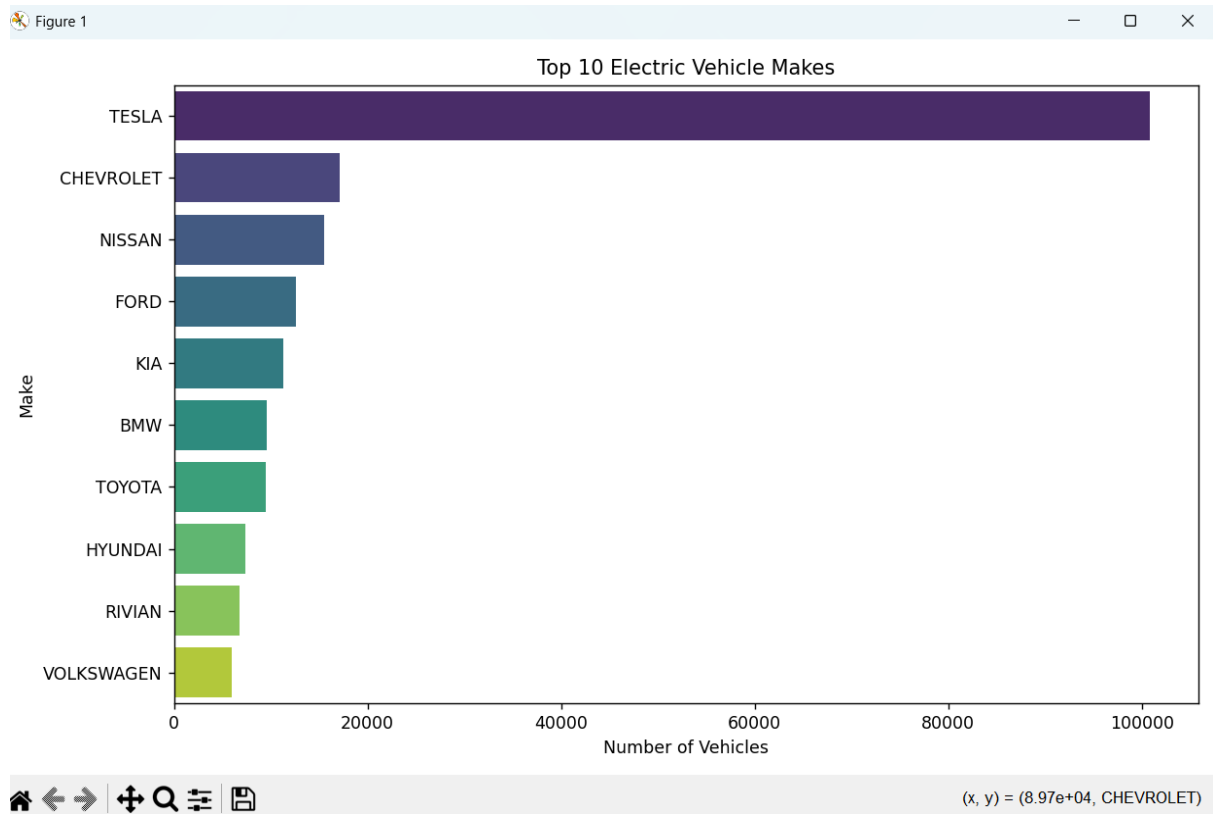
v. Visualization

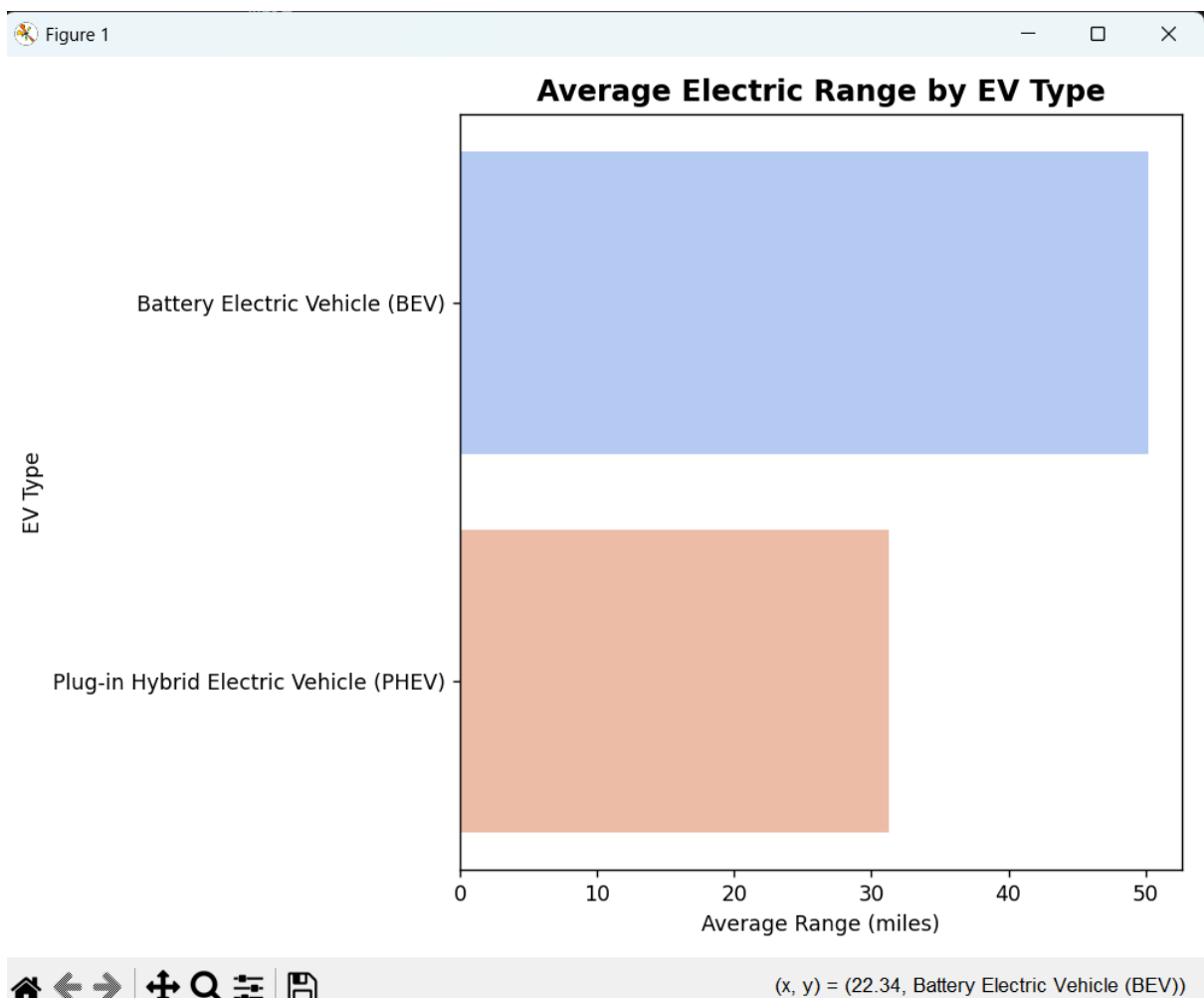
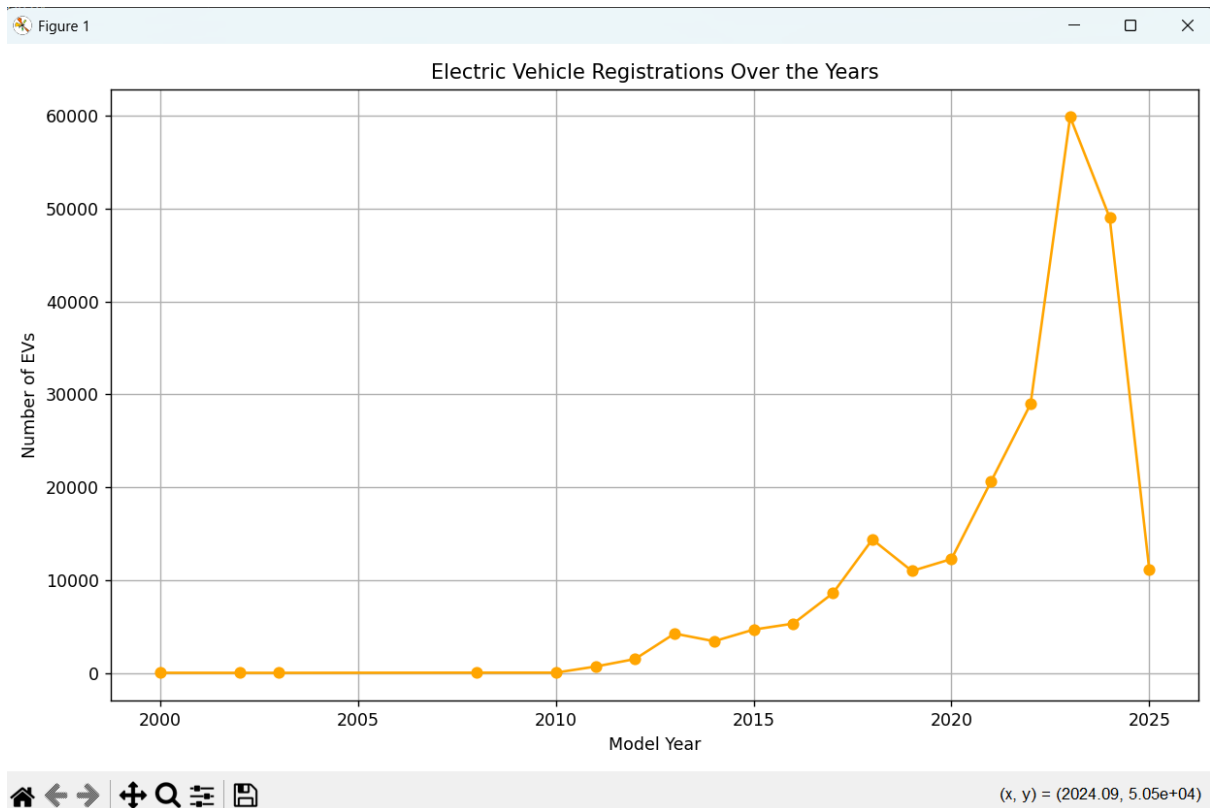
PIE CHART

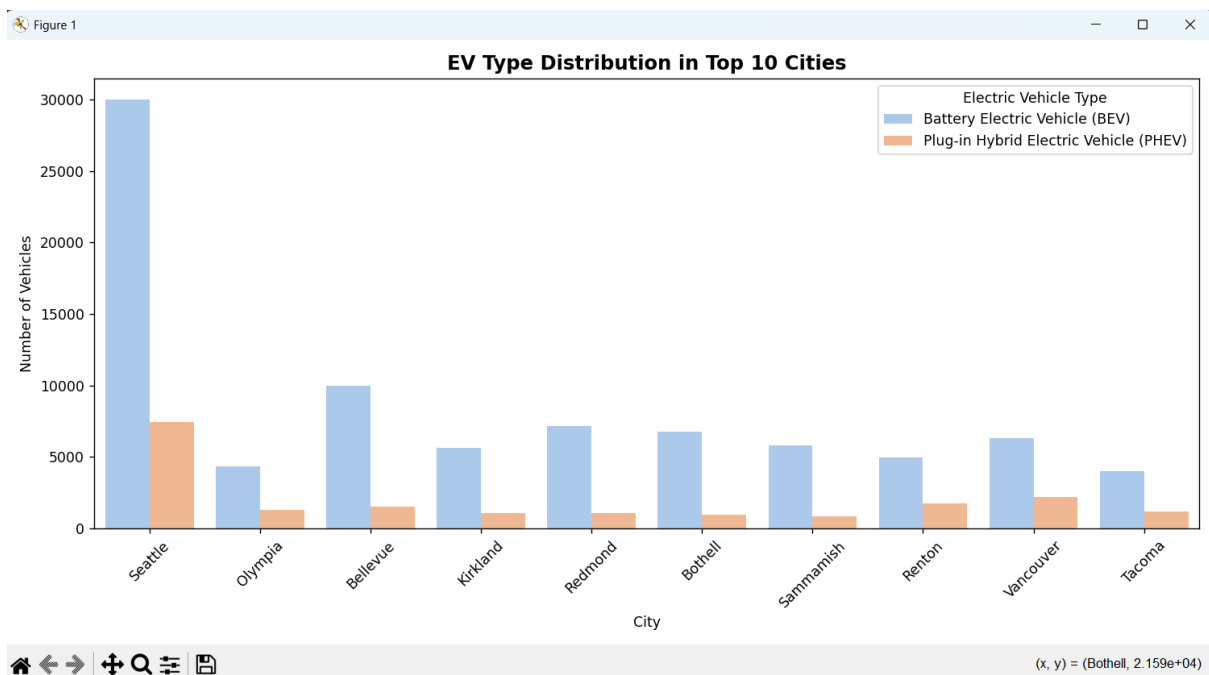
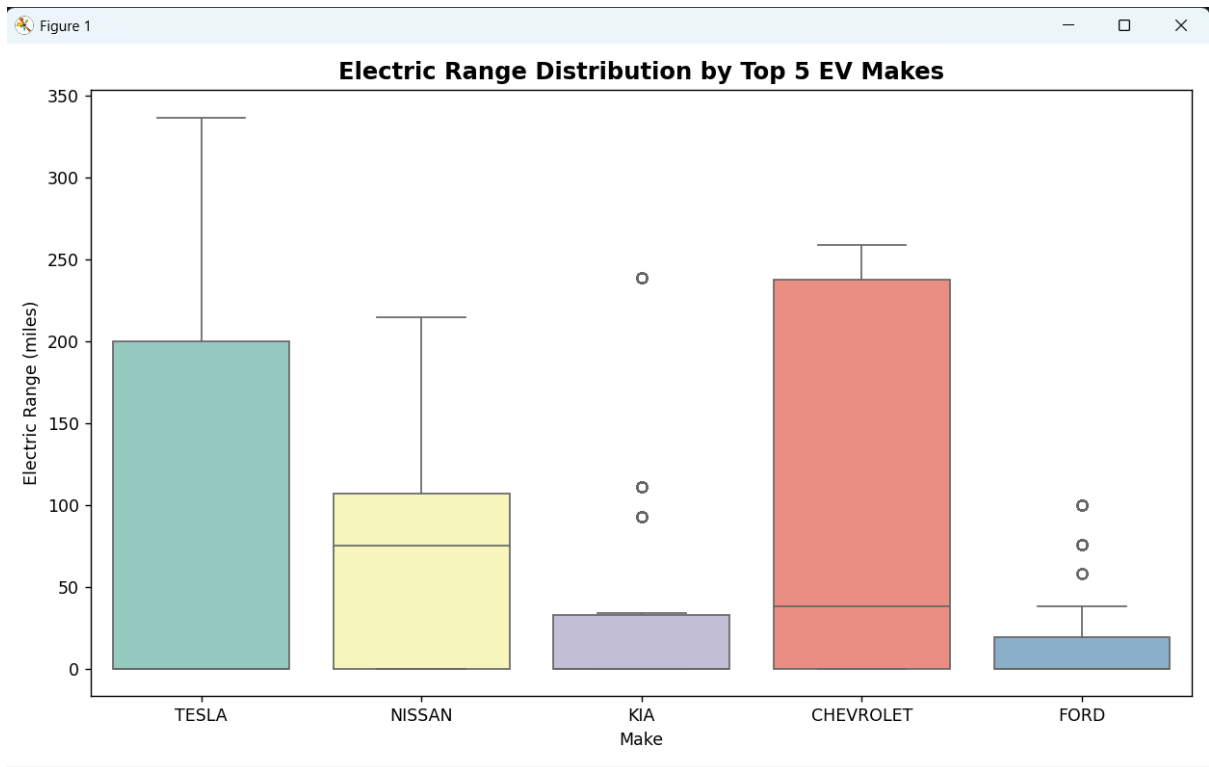


Count chart



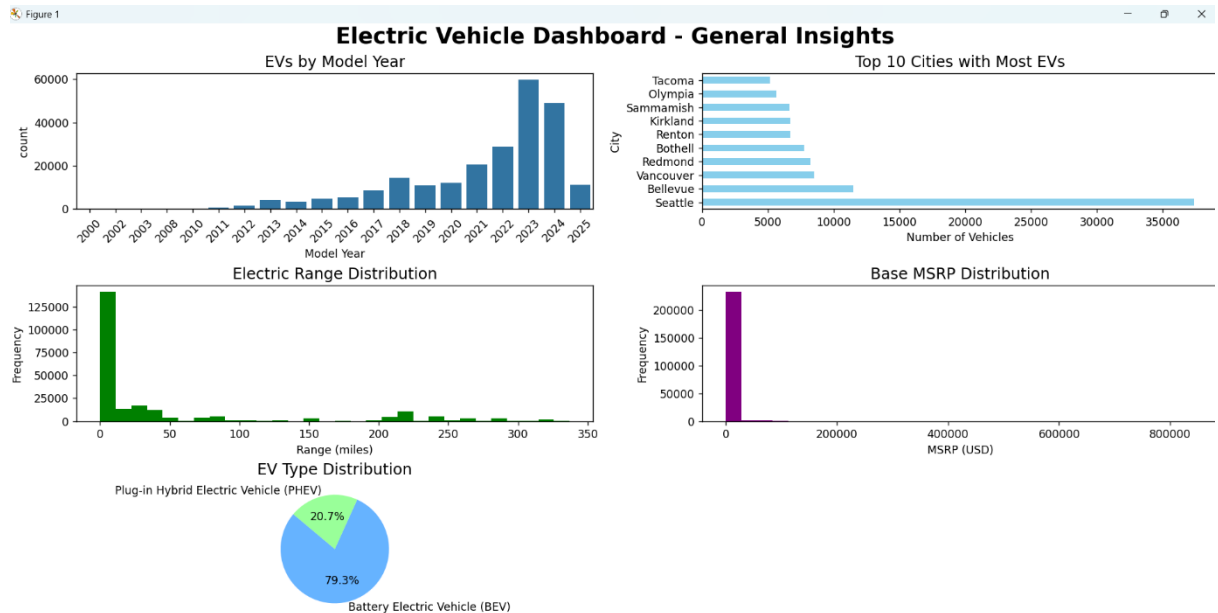
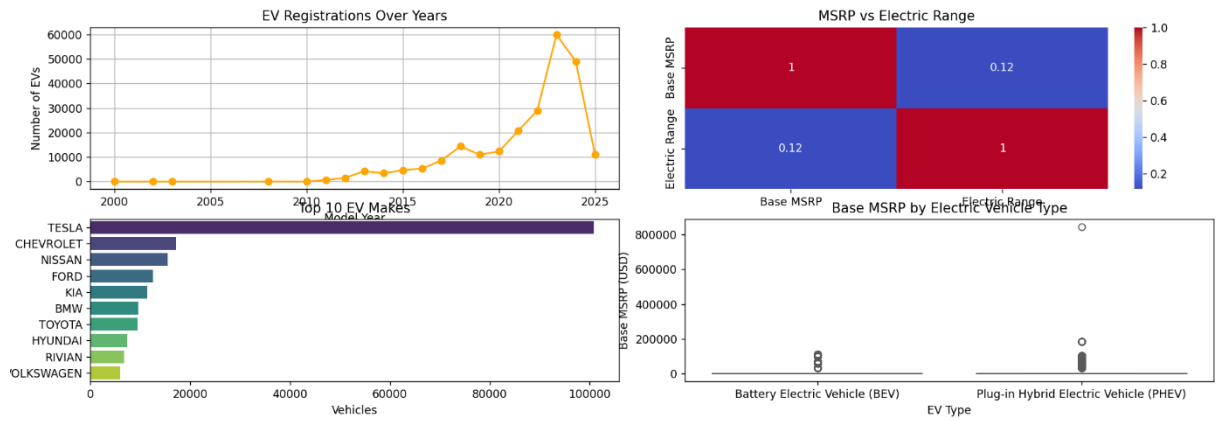






Two dashboards were created:

- **Dashboard 1: General Insights:** 6-subplot grid
- **Dashboard 2: Advanced Trends:** Includes background styling and spacing between charts for visual appeal



Enhancements:

- Improved layout with spacing between subplots
- Added color schemes for aesthetic visualizations
- Background image for better presentation

5. Conclusion

This project successfully demonstrates how data visualization tools can provide actionable insights from complex datasets. The dashboard presents a comprehensive view of EV trends and is a useful tool for stakeholders in the electric vehicle industry.

6. Future Scope

- Integrate real-time EV data APIs
 - Develop an interactive web dashboard using Plotly/Dash
 - Include emissions data and charging infrastructure
 - Machine learning for EV market prediction
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7. References

- <https://pandas.pydata.org/>
- <https://seaborn.pydata.org/>
- <https://matplotlib.org/>
- EV dataset from public government sources

IMPORTANT LINKS

Git hub Link:- <https://github.com/samitkhicher-max/Python-dataset-project-on-Electric-vehicle>

Linkedin Link:-

<https://www.linkedin.com/feed/update/urn:li:activity:7316534235572379648/>

Dataset Link:-

<https://catalog.data.gov/dataset/air-quality>